The Two-Body Problem in the Context of Lagrangians

Kieron Eisma

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Introduction

This is a brief technical note on how one may use the Lagrangian method to simplify the two-body problem, enabling it to be solved as if there were only one particle. As always, email me should you find an incorrect line of working or have suggestions that can make this article more pedagogically sound:

kiwi.eisma@gmail.com

1 Reduced Mass in the Lagrangian

Recall that the (non-relativistic) Lagrangian is defined as

$$\mathcal{L} = T - V,$$

where T and V denote the total kinetic and potential energies of the system, respectively.

Now, consider two particles, masses m_1 and m_2, with respective position vectors $\mathbf{r}_1$ and $\mathbf{r}_2$. We assume that the potential energy of the two objects depends only on their distance.

The Lagrangian is then

$$\mathcal{L} = \frac{1}{2}m_1\dot{\mathbf{r}}_1^2 + \frac{1}{2}m_2\dot{\mathbf{r}}_2^2 - V(|\mathbf{r}_2 - \mathbf{r}_1|). \tag{1}$$

Now, recall that the system's center of mass is given by the position vector

$$\mathbf{r}_{\mathbf{CM}} = \frac{m_1\mathbf{r}_1 + m_2\mathbf{r}_2}{m_1 + m_2}. \tag{2}$$

If we take the origin at the position of the center of mass, that would imply $\mathbf{r_{CM}} = \mathbf{0}$, (see Figure 1) so that we obtain

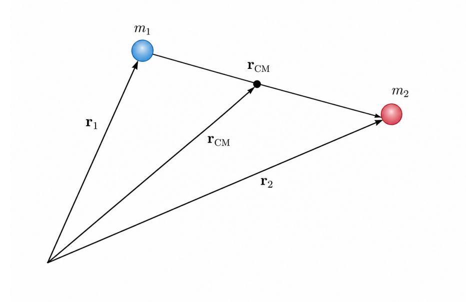

Figure 1: *Two particles of masses m_1 and m_2, and their center of mass. Image generated* by ChatGPT 5.5.

$$m_1\mathbf{r}_1 + m_2\mathbf{r}_2 = \mathbf{0}. \tag{3}$$

from Equation 2.

Moreover, let's define a new vector [1]

$$\mathbf{r} = \mathbf{r}_2 - \mathbf{r}_1. \tag{4}$$

Notice the length of $\mathbf{r}$ is equal to $|\mathbf{r}_2 - \mathbf{r}_1|$.

Now, from Equation 3, we have

$$\mathbf{r}_1 = -\frac{m_2}{m_1}\mathbf{r}_2.$$

Thus,

$$\mathbf{r}_1 = -\frac{m_2}{m_1}(\mathbf{r} + \mathbf{r}_1),$$

$$\begin{aligned}\left(1 + \frac{m_2}{m_1}\right)\mathbf{r}_1 &= \frac{m_1 + m_2}{m_1}\mathbf{r}_1 \\ &= -\frac{m_2}{m_1}\mathbf{r},\end{aligned}$$

[1]You could also define $\mathbf{r} = \mathbf{r}_1 - \mathbf{r}_2$, as in this article.

where the last equality holds by the equation directly above. This gives

$$\mathbf{r}_1 = -\frac{m_2}{m_1 + m_2}\mathbf{r}. \tag{5}$$

By a similar argument, one can show that

$$\mathbf{r}_2 = \frac{m_1}{m_1 + m_2}\mathbf{r}. \tag{6}$$

We substitute Equations 5 and 6 into the Lagrangian given in Equation 1:

$$\begin{aligned}
\mathcal{L} &= \frac{1}{2}m_1\dot{\mathbf{r}}_1^2 + \frac{1}{2}m_2\dot{\mathbf{r}}_2^2 - V(|\mathbf{r}_2 - \mathbf{r}_1|) \\
&= \frac{1}{2}m_1\left(\frac{m_2}{m_1+m_2}\dot{\mathbf{r}}\right)^2 + \frac{1}{2}m_2\left(\frac{m_1}{m_1+m_2}\dot{\mathbf{r}}\right)^2 - V(r) \\
&= \frac{1}{2}\left(\frac{m_1 m_2^2}{(m_1+m_2)^2} + \frac{m_2 m_1^2}{(m_1+m_2)^2}\right)\dot{\mathbf{r}}^2 - V(r) \\
&= \frac{1}{2}\left(\frac{m_1 m_2^2 + m_2 m_1^2}{(m_1+m_2)^2}\right)\dot{\mathbf{r}}^2 - V(r) \\
&= \frac{1}{2}\left(\frac{m_1 m_2(m_1+m_2)}{(m_1+m_2)^2}\right)\dot{\mathbf{r}}^2 - V(r) \\
&= \frac{1}{2}\left(\frac{m_1 m_2}{m_1+m_2}\right)\dot{\mathbf{r}}^2 - V(r) \\
&= \frac{1}{2}\mu\dot{\mathbf{r}}^2 - V(r),
\end{aligned}$$

where $\mu = \frac{m_1 m_2}{m_1+m_2}$ is defined as the *reduced mass* of the system– the "effective mass" of the configuration.

2 Applications

One fun application of the formula above is in the problem of two masses, m_1 and m_2, joined together by a spring of force constant k:

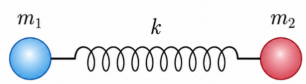

Figure 2: *Two particles of masses m_1 and m_2, connected by a spring of force constant k. Image generated by* ChatGPT 5.5.

We define coordinates so that $\mathbf{r}_1$ and $\mathbf{r}_2$ denote the displacements of each mass from the spring's *equilibrium extension*, allowing us to disregard the spring's length in the Lagrangian.

We use the expression of the Lagrangian in reduced mass, as derived in Section 1, to obtain

$$\mathcal{L} == \frac{1}{2}\mu\dot{\mathbf{r}}^2 - \frac{1}{2}k\mathbf{r}^2,$$

where of course $\mathbf{r} := \mathbf{r}_2 - \mathbf{r}_1$.

Now,

$$\frac{\partial \mathcal{L}}{\partial \dot{\mathbf{r}}} = \mu\dot{\mathbf{r}}, \tag{7}$$

and

$$\frac{\partial \mathcal{L}}{\partial \mathbf{r}} = -k\mathbf{r}. \tag{8}$$

The Euler-Lagrange Equation implies

$$\frac{\partial \mathcal{L}}{\partial \mathbf{r}} - \frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{\mathbf{r}}}\right) = 0,$$

so substituting our findings in Equations 7 and 8 above gives

$$-k\mathbf{r} - \mu\ddot{\mathbf{r}} = 0,$$

or

$$\ddot{\mathbf{r}} + \frac{k}{\mu}\mathbf{r} = 0,$$

which is the familiar harmonic differential equation.[2]

We therefore have

$$\omega = 2\pi f = \sqrt{\frac{k}{\mu}},$$

which yields

$$f = \frac{1}{2\pi}\sqrt{\frac{k}{\mu}}.$$

The equation above is important in IR spectroscopy, a vital analytical tool in chemistry determine the functional groups present in a molecule. For further elaboration, see Chapter 3 of [Cla+12].

3 Conclusion

Using the reduced mass μ of a two-body system we have reduced the original Lagrangian in Equation 1:

$$\mathcal{L} = \frac{1}{2}m_1\dot{\mathbf{r}}_1^2 + \frac{1}{2}m_2\dot{\mathbf{r}}_2^2 - V(|\mathbf{r}_2 - \mathbf{r}_1|).$$

which involves *two* coordinates (one for each particle), into

$$\mathcal{L} = \frac{1}{2}\mu\dot{\mathbf{r}}^2 - V(r)$$

which enables us to tackle the two-body problem as there was only one entity involved.

[2] A more detailed discussion of equations of this form in oscillations is contained in Chapter 4 of [Mor08]. One may also refer to Chapter 1 of [SS03] should they need a more rigorous treatment.

References

[Cla+12] Jonathan Clayden et al. *Organic Chemistry*. 2nd ed. Oxford University Press, 2012.

[Mor08] David Morin. *Introduction to Classical Mechanics: With Problems and Solutions*. Cambridge University Press, 2008.

[SS03] Elias M. Stein and Rami Shakarchi. *Fourier Analysis: An Introduction*. Vol. 1. Princeton Lectures in Analysis. Princeton University Press, 2003.